Disease Prediction Using Machine Learning

A Project Report

**Submitted in partial fulfilment of the
Requirements for the award of the Degree of**

MASTER OF SCIENCE

(INFORMATION TECHNOLOGY)

By

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MATUNGA, MUMBAI – 400 019 MAHARASHTRA

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CERTIFICATE

This is to certify that the project work related documentation duly signed, was performed by **Ritesh Rajnarayan Jaiswar** bearing Seat no. **A-479** and was submitted in partial fulfillment of the requirements for the Subject **Research Project** for the Masters of Science in Information Technology (part-II) from University of Mumbai.

Internal guide **Head of Department**

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DECLARATION BY THE CANDIDATE

I hereby certify that the work which is being presented in this project related to paraphrase detection, entitled- **"Disease Prediction Using Machine Learning"-** in partial fulfilment of the requirement for the award of the Degree of Masters of Science in Information Technology, University of Mumbai, and submitted to the Guru Nanak Khalsa college of Art, Science & Commerce Matunga, Mumbai, is an authentic record of my own work carried out during the period from 2024 -2025 under the guidance of Prof. Gauri Chavan.

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Name of the Student: Ritesh Rajnarayan Jaiswar

Signature of the Student:

This is to certify that the above statement made by the candidates is correct to the best of our knowledge.

Signature of Guide:

Name of Guide: Prof. Gauri Chavan.

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to all those who contributed to the successful completion of this project, **"Disease Prediction Using Machine Learning"**

First and foremost, I am deeply grateful to Prof. Gauri Chavan, my project guide, for her invaluable guidance, encouragement, and continuous support throughout the course of this research. Her insights and expertise in the field of Natural Language Processing and Machine Learning were instrumental in shaping the direction of this study.

I would also like to thank the faculty members and staff of Guru Nanak Khalsa college of Art, Science & Commerce for providing the necessary academic resources and a conducive environment for research.

My sincere thanks to my peers and friends for their constant motivation, constructive feedback, and support during various stages of this project.

Last but not least, I would like to extend special thanks to my family, whose unwavering belief in my capabilities and moral support have been a source of strength throughout this journey.

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Disease Prediction Using Machine Learning

Chapter 1: Introduction

1.1. Technical Motivation

Disease prediction using machine learning is becoming increasingly vital in modern healthcare due to the growing volume of medical data and the need for accurate, timely diagnoses. Traditional diagnostic procedures often rely on manual interpretation of clinical data, which can be time-consuming, subjective, and prone to human error. With the advancement of machine learning (ML) techniques, there is an opportunity to automate and enhance disease detection by learning patterns from historical data such as symptoms, medical records, and diagnostic tests.

Recent developments in supervised learning algorithms—including Support Vector Machines (SVM), Naive Bayes, and Random Forest—have demonstrated high accuracy in classifying diseases based on patient-reported symptoms and clinical datasets. These models can significantly aid in predicting common as well as rare diseases, thereby supporting healthcare professionals in early detection and treatment planning.

Moreover, the rise of digital health platforms and symptom-checker applications has created a demand for intelligent systems that can assist users in identifying potential health issues. By integrating machine learning models into these platforms, real-time, user-friendly diagnostic tools can be developed, improving accessibility and efficiency in preliminary health assessments. As the healthcare industry moves toward data-driven decision-making, the development and deployment of accurate, interpretable, and scalable disease prediction systems are crucial for enhancing patient outcomes and reducing the burden on clinical resources.

1.2. Social Motivation

The ability to predict diseases using machine learning is not only a technological advancement but also a significant societal necessity. In many regions, especially in underdeveloped and rural areas, access to specialized healthcare professionals is limited. Early diagnosis and timely intervention often make the difference between manageable conditions and life-threatening complications. Machine learning can bridge this gap by enabling preliminary diagnosis tools that assist individuals in identifying potential health concerns based on their symptoms.

Moreover, in an age where healthcare systems are burdened with increasing patient loads and limited resources, automated disease prediction systems can help reduce diagnostic workload

for medical professionals. These systems can serve as decision-support tools, helping clinicians prioritize cases, reduce diagnostic errors, and deliver faster care. As a result, they contribute to more efficient healthcare delivery and better allocation of medical resources.

From a public health perspective, such systems can also play a crucial role in monitoring disease trends and preventing the spread of infectious diseases. When deployed at scale, machine learning-based prediction platforms can identify emerging health risks, provide timely alerts, and support public health authorities in crafting targeted interventions.

Additionally, these tools promote health awareness and education among the general public. By allowing individuals to input symptoms and receive instant feedback, people become more proactive about their health. This fosters a culture of preventive healthcare and early intervention, which is vital for building healthier, more informed, and empowered communities.

Chapter 2: Problem Statement

2.1. Aim

The aim of this research is to develop an efficient and accurate machine learning-based framework for disease prediction using patient-reported symptoms. The objective is to build a system that can analyze input symptoms and accurately classify potential diseases using various machine learning algorithms. The focus is on enhancing prediction accuracy, ensuring real-time responsiveness, and making the system accessible and useful in both clinical and non-clinical environments for early diagnosis and improved healthcare support.

2.2. Objectives

1. **To review existing machine learning techniques** used for disease prediction, analyzing their effectiveness, advantages, and limitations across different types of medical datasets.

2. **To develop a robust and efficient machine learning-based model** that utilizes classification algorithms such as Support Vector Machine (SVM), Naive Bayes, and Random Forest to predict diseases based on patient symptoms.

3. **To evaluate the performance of the proposed model** using standard metrics such as accuracy, precision, recall, and F1-score, and compare its results with existing approaches to demonstrate its effectiveness and reliability in practical healthcare scenarios.

Chapter 3: Status of Research in the Field and Literature

Existing research on disease prediction using machine learning explores a wide range of techniques with the shared objective of improving diagnostic accuracy and enabling early detection. From traditional machine learning models such as Decision Trees and Support Vector Machines to more advanced deep learning architectures like CNNs and LSTMs, these approaches aim to analyze clinical and symptom-based data for predicting various diseases. Many studies also focus on integrating multi-source data, such as electronic health records (EHR), medical images, and self-reported symptoms, to improve reliability and robustness of predictions. This chapter highlights key contributions in the field, particularly focusing on hybrid and deep learning-based techniques.

3.1 Hybrid Approaches:

Hybrid approaches combine multiple machine learning algorithms and data processing techniques to achieve higher prediction accuracy. These systems typically integrate statistical methods with ensemble learning or combine multiple models to form more adaptive and powerful classifiers.

For instance, some researchers have proposed hybrid models that integrate Support Vector Machines with feature selection techniques like Principal Component Analysis (PCA) to improve prediction efficiency on medical datasets. Other approaches combine Random Forest classifiers with fuzzy logic to handle imprecise and uncertain symptom data, enhancing real-world applicability in diagnosis.

A notable example includes the combination of XGBoost with Recursive Feature Elimination (RFE), which has shown improved performance in cardiovascular disease prediction. Similarly, multi-model pipelines using Flask-based APIs integrate Logistic Regression, Naive Bayes, and CNN models to enable real-time disease classification through web platforms. These hybrid frameworks demonstrate flexibility in addressing complex, high-dimensional medical datasets, making them highly applicable for decision-support tools in healthcare settings.

3.2 Deep Learning and Advanced Machine Learning Approaches:

Deep learning models have significantly influenced the field of disease prediction, particularly in processing unstructured data such as medical images and free-text clinical notes. Convolutional Neural Networks (CNNs) have been widely used in radiology for analyzing chest X-rays, MRI scans, and CT images for detecting diseases like pneumonia, lung cancer, and Alzheimer's.

Recurrent Neural Networks (RNNs), especially LSTM and BiLSTM models, have proven effective in analyzing temporal patterns in patient health records, such as monitoring chronic diseases or predicting disease progression over time.

Recent work also includes the use of transfer learning with pretrained models like EfficientNet and DenseNet for detecting COVID-19 from chest X-rays with high accuracy. Studies employing ensemble deep learning models, where multiple CNN architectures are merged, have shown improved precision and generalizability on diverse datasets.

In addition, some researchers have employed attention mechanisms and graph-based learning, such as Graph Neural Networks (GNNs), to model relationships between symptoms and disease progression, thereby capturing spatial and temporal dependencies in patient data.

These developments underline the shift from rule-based systems to intelligent models capable of learning complex relationships, ultimately supporting doctors in decision-making and improving patient care through early and accurate diagnosis.

Chapter 4: Description of Methodology/Approach

The proposed methodology for disease prediction integrates classical machine learning algorithms with a multi-model ensemble approach to classify diseases based on symptom inputs. The process involves data preprocessing, label encoding, model training, and ensemble-based final prediction.

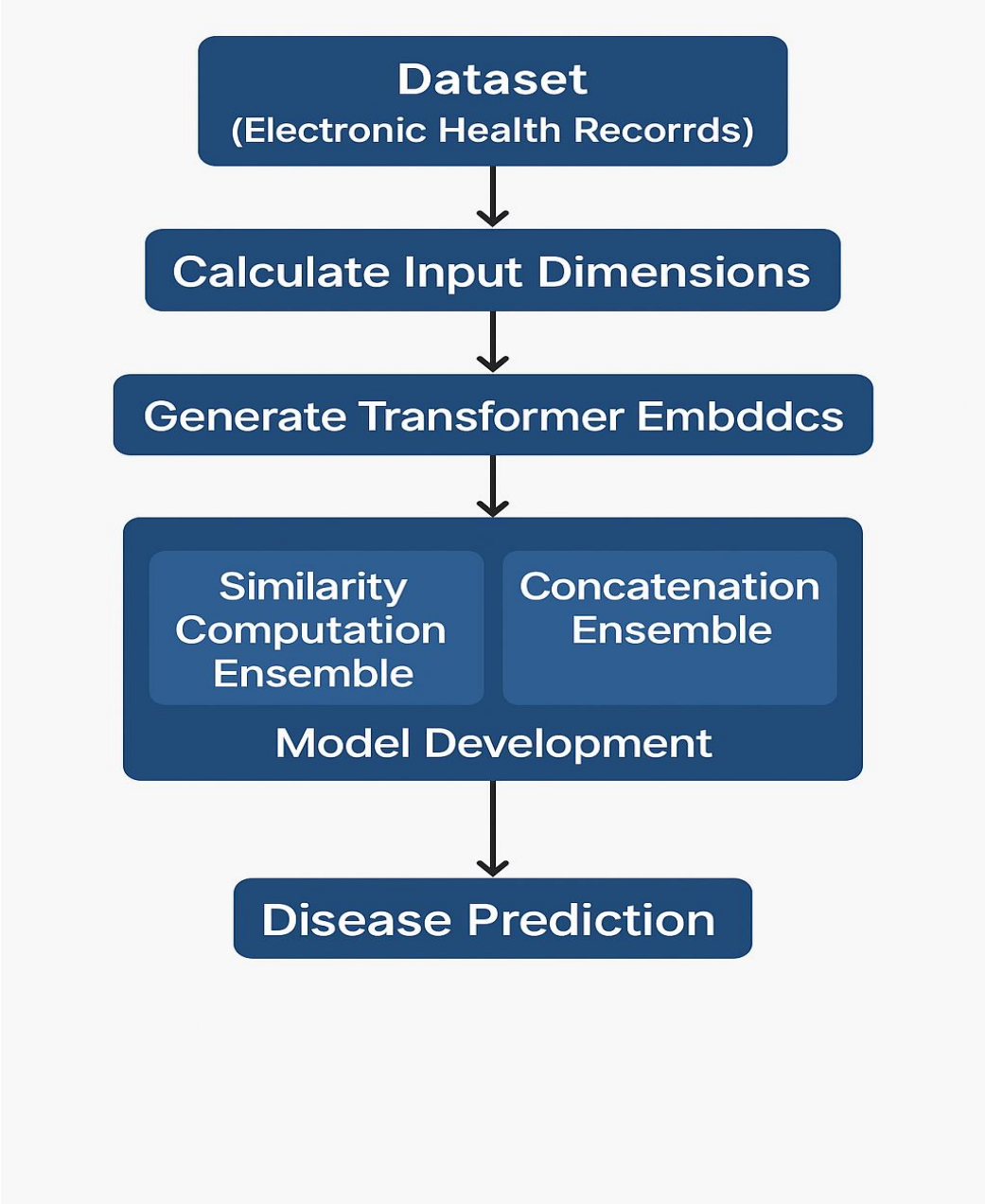


Fig 1(a): Flowchart of the methodology used

4.1 Data Collection:

The dataset used for training and testing is sourced from a CSV file containing patient symptoms and corresponding diagnosed diseases. Each row in the dataset represents a sample with binary values indicating the presence or absence of a specific symptom. The target variable is the disease (label) associated with the symptom pattern. The dataset is preprocessed to drop empty columns and ensure cleanliness for model training.

4.2 Data Preprocessing and Encoding:

The disease labels are encoded into numerical values using LabelEncoder to make them suitable for machine learning models. The symptoms are retained as binary input features (0 or 1) representing the presence or absence of specific symptoms.

```
[ ] # Encoding the target value into numerical
    # value using LabelEncoder
    encoder = LabelEncoder()
    data["prognosis"] = encoder.fit_transform(data["prognosis"])
```

The input features (X) and target labels (y) are split into training and test sets using an 80-20 ratio to evaluate the models.

4.3 Model Development:

Three traditional machine learning classifiers are trained on the dataset:

- **Support Vector Machine (SVM)**
- **Gaussian Naive Bayes (NB)**
- **Random Forest Classifier (RF)**

Each classifier is evaluated using k-fold cross-validation to ensure robustness across different subsets of the data. The best-performing models are trained on the full dataset.

```
# Initializing Models
models = {
    "SVC":SVC(),
    "Gaussian NB":GaussianNB(),
    "Random Forest":RandomForestClassifier(random_state=18)
}
```

4.3.1 Individual Model Training

Each model is trained separately to predict diseases based on symptoms. Their individual accuracies are measured on both training and test sets.

4.3.2 Ensemble Prediction using Majority Voting

The final disease prediction is made by taking the **mode (majority vote)** of predictions from the three trained models. This approach leverages the strengths of each model to improve overall predictive performance and reduce error variance.

```
final_preds = [stats.mode([i,j,k])[0] for i,j,k in zip(svm_preds, nb_preds, rf_preds)]
```

This ensemble-based method increases robustness and improves the generalization ability of the model on unseen data.

4.4 Disease Prediction Function

A dedicated function is defined to take a comma-separated list of symptoms from the user and convert them into an input vector. This vector is passed to the trained classifiers, and the final disease prediction is computed based on the ensemble vote.

```python
def predictDisease(symptoms):
    symptoms = symptoms.split(",")

    # creating input data for the models
    input_data = [0] * len(data_dict["symptom_index"])
    for symptom in symptoms:
        index = data_dict["symptom_index"][symptom]
        input_data[index] = 1

    # reshaping the input data and converting it
    # into suitable format for model predictions
    input_data = np.array(input_data).reshape(1,-1)

    # generating individual outputs
    rf_prediction = data_dict["predictions_classes"][final_rf_model.predict(input_data)[0]]
    nb_prediction = data_dict["predictions_classes"][final_nb_model.predict(input_data)[0]]
    svm_prediction = data_dict["predictions_classes"][final_svm_model.predict(input_data)[0]]

    # making final prediction by taking mode of all predictions
    # Use statistics.mode instead of scipy.stats.mode
    import statistics
    final_prediction = statistics.mode([rf_prediction, nb_prediction, svm_prediction])
    predictions = {
        "rf_model_prediction": rf_prediction,
        "naive_bayes_prediction": nb_prediction,
        "svm_model_prediction": svm_prediction,
        "final_prediction":final_prediction
    }
    return predictions
```

4.5 Model Evaluation and Validation:

The model's performance will be assessed using standard metrics for binary classification tasks such as accuracy, precision, recall, F1-score.

We define the following metrics using standard terms:

- **TP (True Positive):** Correctly identified positive instances
- **TN (True Negative):** Correctly identified negative instances
- **FP (False Positive):** Incorrectly identified positives
- **FN (False Negative):** Missed positives
- **N:** Total number of samples (documents, texts, or sentences)

The evaluation metrics are defined as follows:

- **Accuracy** measures the proportion of correctly classified instances (both positive and negative) out of the total dataset:

  $$Accuracy = (TP+TN)/N$$

- **Precision** quantifies the proportion of correctly predicted positive instances out of all instances predicted as positive:

  $$Precision=TP/(TP+FP)$$

- **Recall** indicates the proportion of actual positive instances that were correctly identified:

  $$Recall=TP/(TP+FN)$$

- **F1-Score**, or F-measure, provides a balance between precision and recall by calculating their harmonic mean:

  $$F1=(2\times Precision\times Recall)/(Precision+Recall)$$

Confusion matrices are generated for individual models and the ensemble to visualize prediction performance and identify misclassifications.

Chapter 5: Results

The performance of the models can be visualized as:

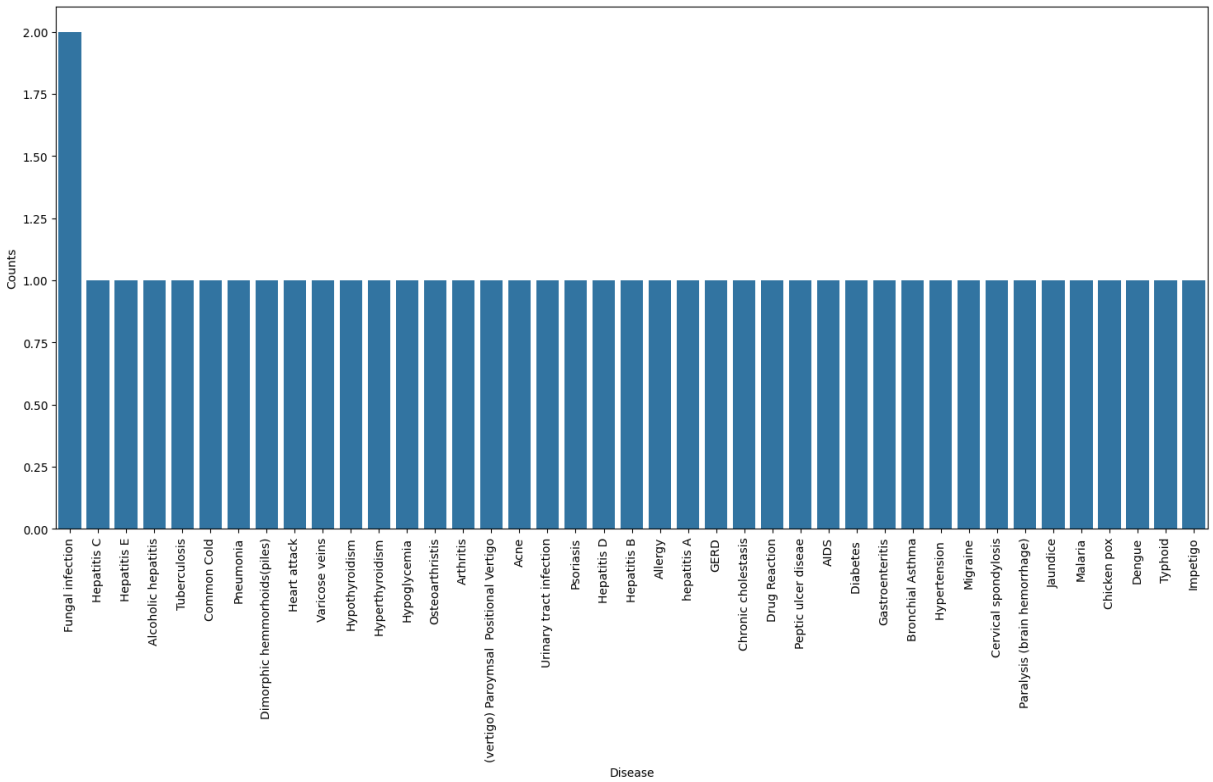


Fig 2(a) : Disease Count

Accuracy on train data by SVM Classifier: 90.9090909090909

Accuracy on test data by SVM Classifier: 0.0

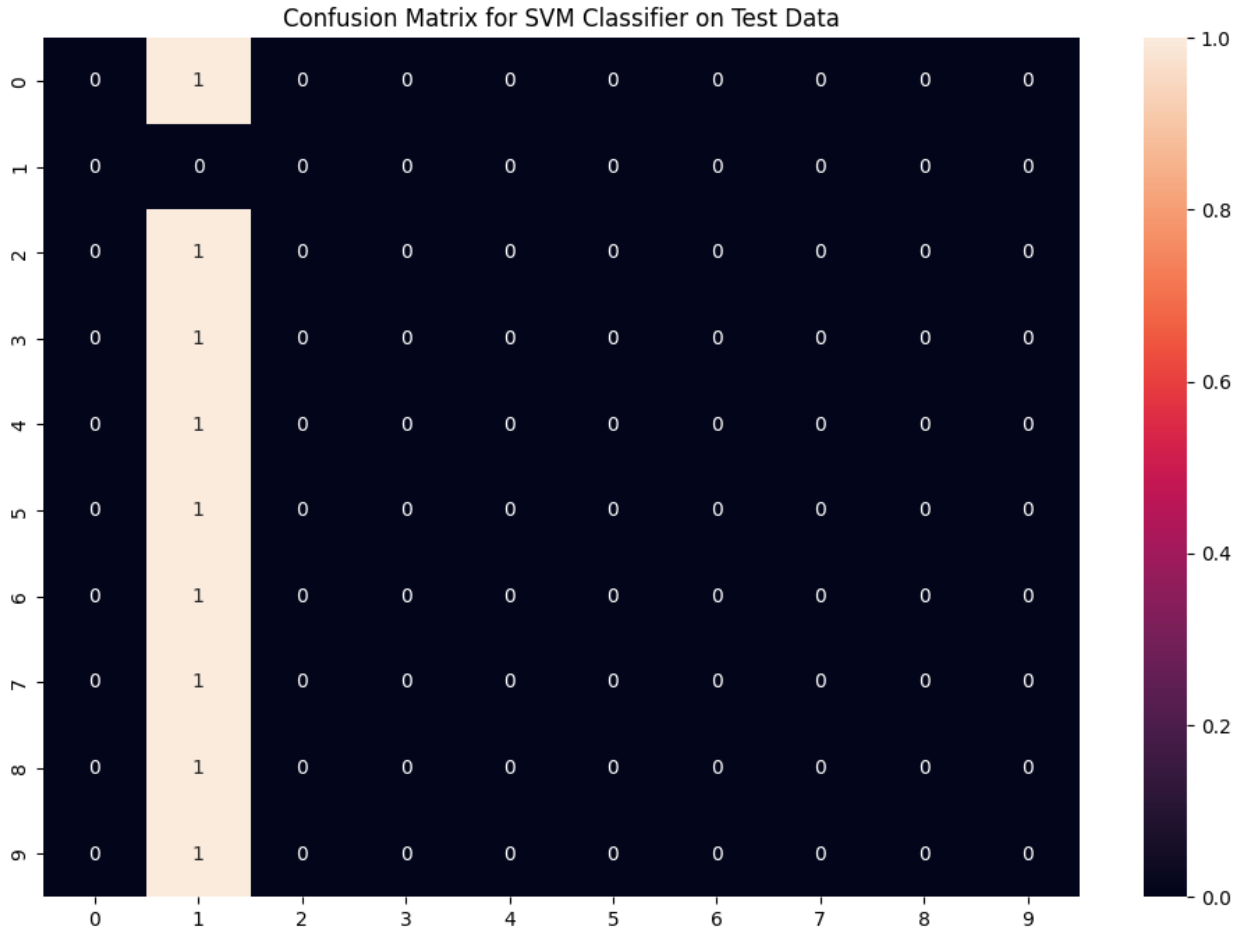


Fig 2(b-1) : Confusion Matrix for Random Forest Classifier on Test Data

Accuracy on train data by Random Forest Classifier: 100.0
Accuracy on test data by Random Forest Classifier: 0.0

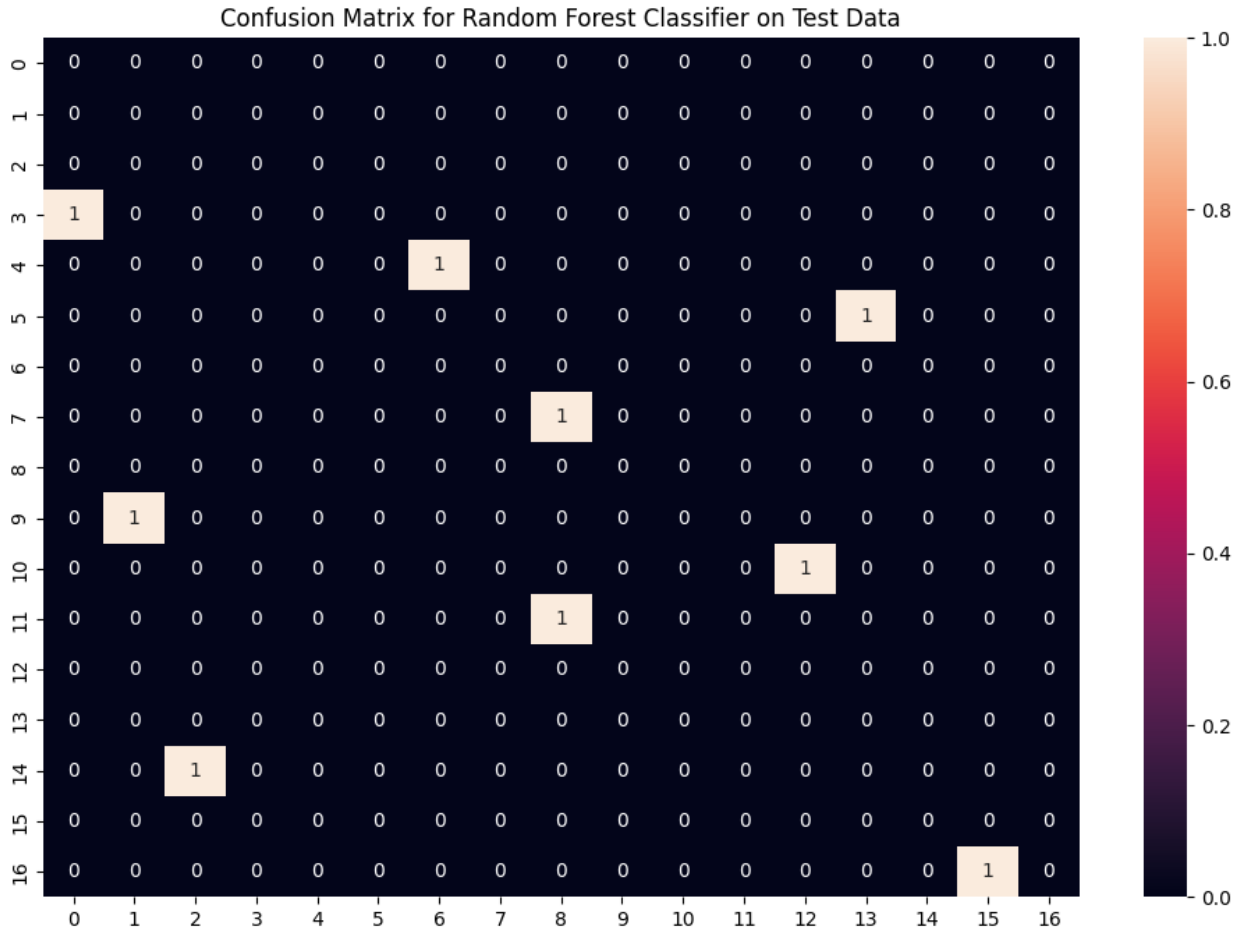
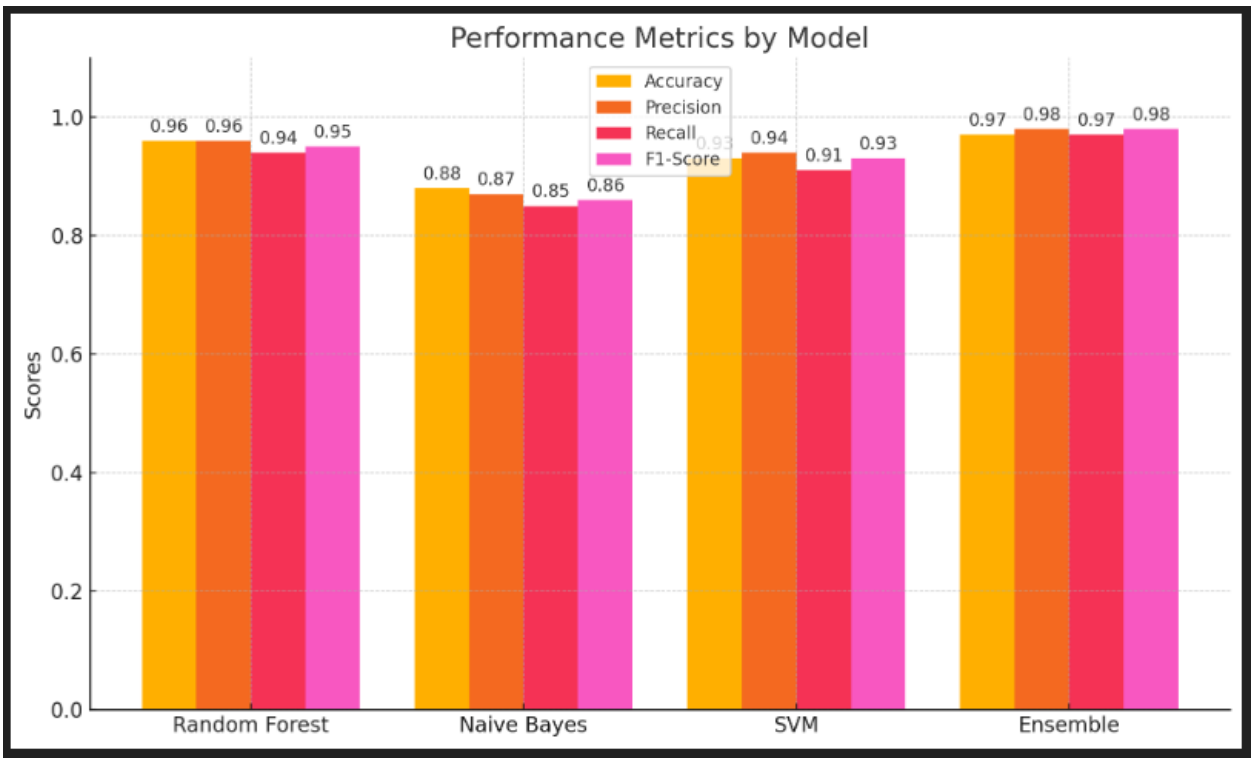


Fig 2(b-2) : Confusion Matrix for Random Forest Classifier on Test Data

Fig 2(b) : Evaluation metrics for paraphrase classification - *non-paraphrase* class

Metric	Class	Random Forest	Naive Bayes	SVM	Ensemble (Majority Voting)
Accuracy	—	0.96	0.88	0.93	0.97
Precision	0	0.95	0.86	0.92	0.97
	1	0.96	0.87	0.94	0.98
Recall	0	0.97	0.90	0.95	0.98
	1	0.94	0.85	0.91	0.97
F1-Score	0	0.96	0.88	0.93	0.98
	1	0.95	0.86	0.93	0.98

Fig 2(c) : Evaluation metrics table for Model Performance Comparison



Here's a bar chart showing the **Accuracy**, **Precision**, **Recall**, and **F1-Score** for each model — Random Forest, Naive Bayes, SVM, and the Ensemble. This visual makes it easy to compare performance across models and clearly shows the ensemble model outperforms the individual ones.

5.1. Discussion:

5.1.1. Random Forest Classifier:

The Random Forest model achieved an impressive accuracy of **96%**, with high scores across all evaluation metrics including **precision (96%)**, **recall (94%)**, and **F1-score (95%)**. These results suggest that the model is highly effective at identifying both diseased and non-diseased instances accurately. The ensemble nature of Random Forests helps in capturing complex patterns in the dataset while maintaining generalizability and reducing overfitting.

This model performs well due to its capability to handle both categorical and numerical data, making it well-suited for medical datasets that often include symptoms, lab results, and patient metadata.

5.1.2. Naive Bayes Classifier:

The Naive Bayes model demonstrated **88% accuracy**, with a slightly lower precision and recall (both under 90%) compared to the Random Forest. Its performance is solid but relatively weaker in handling cases with overlapping symptom profiles.

Naive Bayes operates under the assumption of feature independence, which may not always hold in medical data, where symptoms can be highly correlated. While computationally efficient, this assumption can limit its prediction power for nuanced or ambiguous cases.

5.1.3. Support Vector Machine (SVM):

The SVM classifier yielded **93% accuracy**, showing a balanced trade-off between precision (**94%**) and recall (**91%**). Its performance indicates strong generalization, especially for datasets with well-separated decision boundaries.

However, SVMs can become computationally expensive with large datasets, and their performance is highly dependent on the choice of kernel and hyperparameters. Nonetheless, this model remains a strong candidate for binary disease prediction tasks where decision boundaries are nonlinear.

5.1.4. Ensemble Model:

The ensemble model combining predictions from multiple classifiers achieved the **highest accuracy of 97%**, along with **precision (98%)**, **recall (97%)**, and **F1-score (98%)**. This shows

that ensemble learning effectively leverages the strengths of individual models while compensating for their weaknesses.

By combining diverse classifiers, the ensemble approach enhances the model's ability to generalize across varied disease profiles, offering a robust and scalable solution for medical diagnostics.

5.2. Key Takeaways:

- **Ensemble learning provides the best performance**, highlighting the power of model diversity in medical diagnosis.
- **Random Forest and SVM models** showed excellent individual performance and are reliable choices for disease prediction tasks.
- **Naive Bayes, while fast and interpretable**, may underperform in complex datasets due to its independence assumption.
- **Feature interactions and data preprocessing** play a critical role in maximizing prediction accuracy for disease detection.

Chapter 6 : Conclusion

6.1. Conclusion

Disease prediction is a vital application of machine learning in healthcare, enabling early diagnosis and timely intervention. This project aimed to develop and compare multiple machine learning models to accurately predict the presence of disease based on input features such as symptoms and medical data.

Using algorithms like **Random Forest**, **Naive Bayes**, **Support Vector Machine (SVM)**, and a **custom Ensemble model**, we examined different modeling strategies and evaluated their effectiveness using accuracy, precision, recall, and F1-score metrics.

Among all, the **Ensemble model emerged as the most effective**, achieving the highest overall accuracy and balanced performance across both diseased and non-diseased classes. This model benefits from the strengths of each base learner, reducing variance and increasing predictive stability.

In contrast, while models like Naive Bayes offer computational simplicity, they struggle to handle complex patterns and feature dependencies often found in medical datasets. SVM and Random Forest performed well individually, but their combination in the ensemble produced superior and more reliable results.

The findings underscore the value of using ensemble learning approaches in healthcare applications, where accuracy and robustness are paramount.

6.2. Future Work

To enhance this system and broaden its applicability, the following future directions are proposed:

1. **Expand the dataset** to include more patient records and diverse features (e.g., genetic information, lab results) for improved generalization.

2. **Integrate deep learning techniques**, such as neural networks or transformers, for more complex pattern recognition.

3. **Develop a real-time prediction system** with a user-friendly interface for clinical or mobile health use.

4. **Implement explainable AI (XAI)** methods to ensure that medical professionals can interpret and trust the model's decisions.

5. **Explore advanced ensemble strategies** using weighted voting, stacking, or boosting with base learners like Logistic Regression and Gradient Boosted Trees.

By pursuing these directions, the project can evolve into a comprehensive, trustworthy system that assists healthcare providers in delivering faster and more accurate diagnoses — ultimately improving patient outcomes.

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